BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI, Hyderabad Campus,
Second Semester, 2024-2025, Analog Electronics (EEE/ ECE/INSTR F341)
Comprehensive Examination (Closed Book)

Time: 2.00 to 5.00 pm (3 Hours) Date: 03-05-2025 Max. Marks: 80 (40+40)

PART-A

Name **ID No.**

General Instructions

1. The question paper has two parts: **Part A and Part B.**
2. For Part A, answers should be written in Part-A Question cum answer booklet.
3. **3.** Maximum time for PART A is **90 minutes from the start of the exam (2.00 to 3.30 pm).**
4. **If you feel none of the given options are correct for any of the question in PART A, Write E in the box (E indicates none of the given options).**
5. Upon submission of Part-A Question paper cum answer booklet, Part-B question paper cum answer booklet will be provided. Early submission of PART-A (before 3.30 pm) is allowed
6. For Part B, answers should be written in the Part-B question paper cum answer booklet.
7. Only **one** question paper cum answer sheet for each part will be provided to each student.
8. The **main answer booklet is for your rough work** only, and it will not be evaluated.
9. Answers written with **pencil** will not be evaluated.
10. Exchange of calculators/pens/pencils/erasers or **usage of white ink** is strictly prohibited. If **white ink** used for any answer, it will not be evaluated.
11. You shall be allowed to use the **washroom only once between 3.40 to 4.40 pm.**
12. There is **no negative marking** for any question in Part A and Part B.
13. Usage of any electronic gadgets except a calculator is strictly prohibited.
14. Unfair means will lead to severe consequences as per the rules of the Institute.

Specific Instructions for Part-A

1. Attempt all the questions. Each question has only one correct option and carries 2 marks. **Marks: 20 X 2 = 40**
2. Write the correct option i.e. **(A), (B), (C), or (D)** in the space provided. For over-writing / ambiguous writing /options written using lower case letters zero marks will be awarded.

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1		**6**		**11**		**16**	
2		**7**		**12**		**17**	
3		**8**		**13**		**18**	
4		**9**		**14**		**19**	
5		**10**		**15**		**20**	

Part A Marks: ______ **/40**

1. A 6-bit dual-slope ADC has reference voltage V_{REF} = 2.5 V and operated at clock frequency is 1 MHz. The digital equivalent of an input voltage V_{IN} = 1.5625 V and the corresponding conversion time (neglecting switching delay, if any) will be

A. 100100 and 60 µs **B. 101000 and 104 µs** C. 111001 and 57 µs D.011101 and 69 µs

2. In a 4-bit Successive approximation type ADC, the MSB is stuck at zero due to some malfunctioning. If full scale voltage is 15 V, determine the digital output from ADC for an input of 10 V.

A. 0111 B.1010 C. 0101 D. 1000

3. For V_R = 5 V and R = 4 kΩ, current 'i' through the resistor 2R connected to ground as shown in **Fig. 1** is

A. 31.25 µA B. 62.5 µA **C. 78.12 µA** D. 156.25 µA

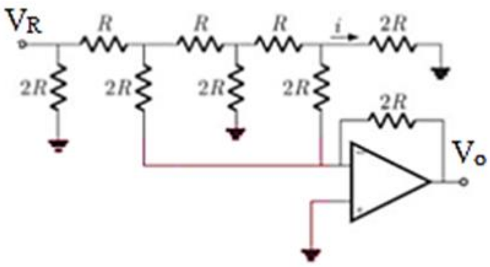


Fig. 1

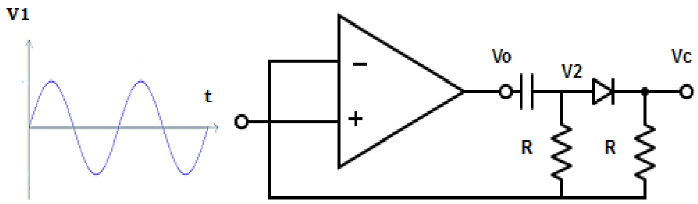
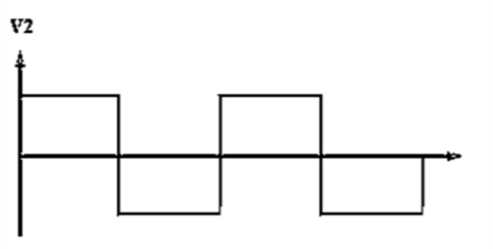
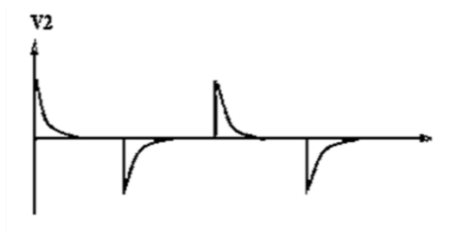
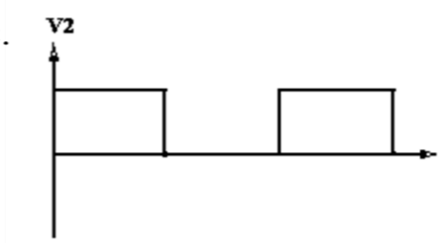


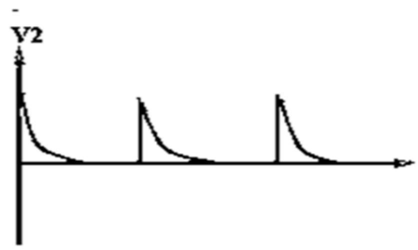
Fig. 2

4. An opamp circuit and its input are shown in **Fig. 2**. Choose the correct output voltage waveform at V_2.
Ans: B

A.


B.


C.


D.


5. Three identical amplifiers each having a gain of $Ao/2 \angle 60^{\circ}$ are connected in cascade. The positive feedback loop has again of 0.008. the value of Ao that will render the cascaded system oscillatory is

A. 10 B. -10 C. 250/3 D. 250

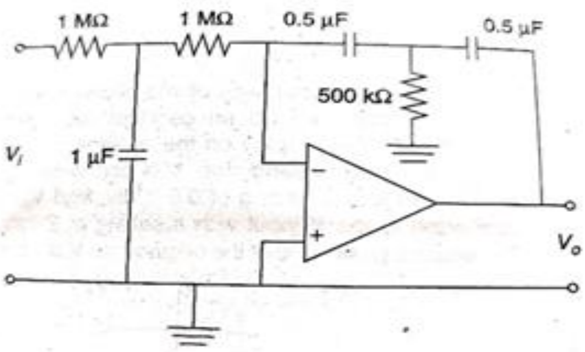


Fig. 3

6. The ideal opamp circuit shown in **Fig. 3** acts as a

A. Low pass filter B. High pass filter **C. Band pass filter** D. Band reject filter

7. In the LPF circuit shown in **Fig. 4**, the switches (S) are operated (On/Off) at a clock frequency of 15.7 kHz. The opamp has unity-gain bandwidth of 1 MHz. Determine the 3-dB bandwidth of the LPF circuit.

A. 1 MHz **B. 5 kHz** C. 15.7 kHz D. $1/\sqrt{2}$ MHz

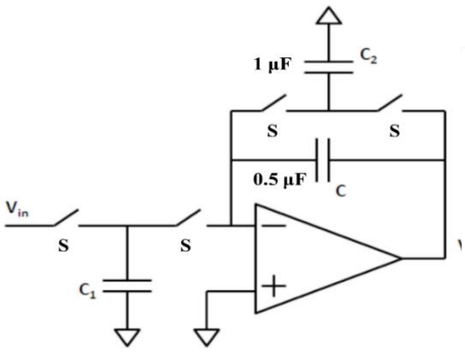


Fig. 4

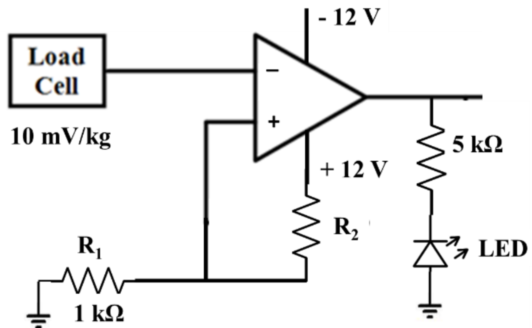


Fig. 5

8. The circuit shown in **Fig. 5** is used in elevators to indicate passenger overweight. Assume the maximum weight allowed in the elevators is 680 kg. The LED will glow to indicate overweight. The opamp has saturation voltage of ±10 V and the weight-sensor (load cell) generate 10 mV voltage per kg weight. If R_1 = 1 kΩ, determine the value of R_2.

A. 784 Ω **B. 764 Ω** C. 820 Ω D. 826.6 Ω

9. A particular opamp has a slew rate of 0.5 V/μs. It is used as a non-inverting amplifier with a gain of 25. The voltage gain against frequency curve is flat up to 50 kHz. Calculate the maximum peak to peak input signal that can be applied to get the undistorted output.

A. 63.6 mV B. 1.59 V **C. 127.3 mV** D. 3.18 V

10. The analog input voltage ranges from -5 V to +8 V in a 9-bit ADC, what is the binary output when the analog input is zero volts.

A. 011001101 **B. 011000101** C. 111001101 D.111000101

11. A DAC-based ADC uses an 8 – bit DAC. MSB of the DAC output is 6V, Vin = 7.2 V and system clock is 100 kHz. What will be the DAC output when the system clock runs for 1.2 ms.

A.6 V B. 1.2 V **C. 5.6 V** D. 7.2 V

12. Find out duty cycle for the circuit shown in **Fig. 6**. Assume a sinusoidal input of peak-to-peak Voltage, V_{pp}

12 V is applied, V_{cc}=5V and opamp V_{sat} = ± 13V, Zener diode has V_Z = 3V and R_1 = 10kΩ.

A. 55 %   **B. 66.67%**   C. 33.33%   D. 75%

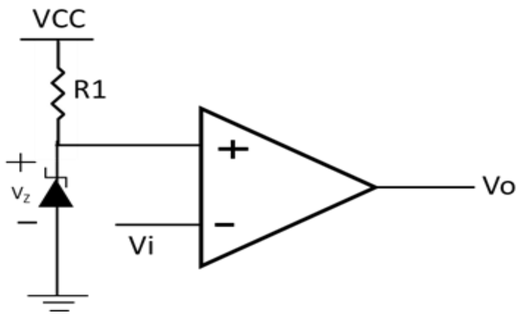


Fig. 6

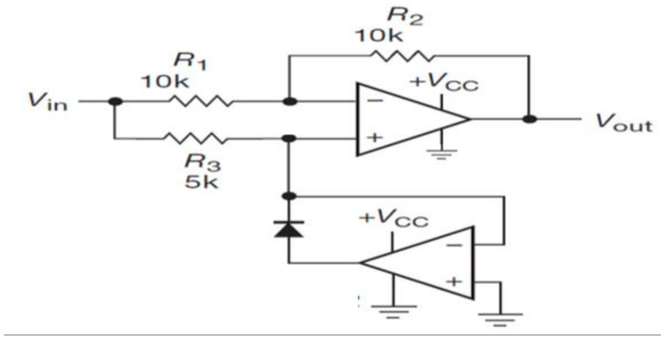
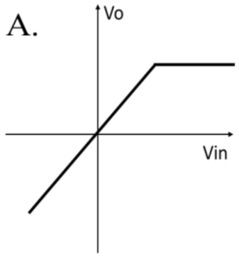


Fig. 7

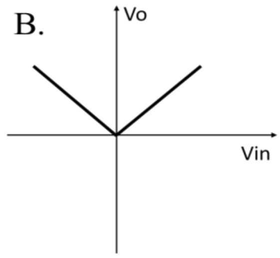
13. Assume the diode to be ideal in **Fig. 7**. Determine the transfer characteristic of the circuit.

Ans: B

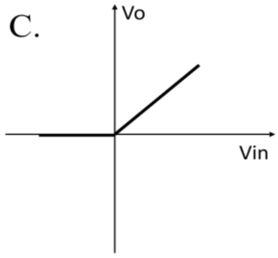
A.



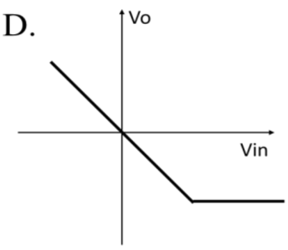
B.



C.



D.



14. Fnd V_{out} in the circuit shown in **Fig. 8**. Given V_x = 2V and V_y = 4V, V_{CC} = 12 V and V_{EE} = -12 V, and R_1 = R_2 and R_3 = R_4. Assume both the transistors are identical and thermal voltage V_T = 26 mV.

A. 18.8 mV   B. 26 mV   **C. 18 mV**   D. None of the options

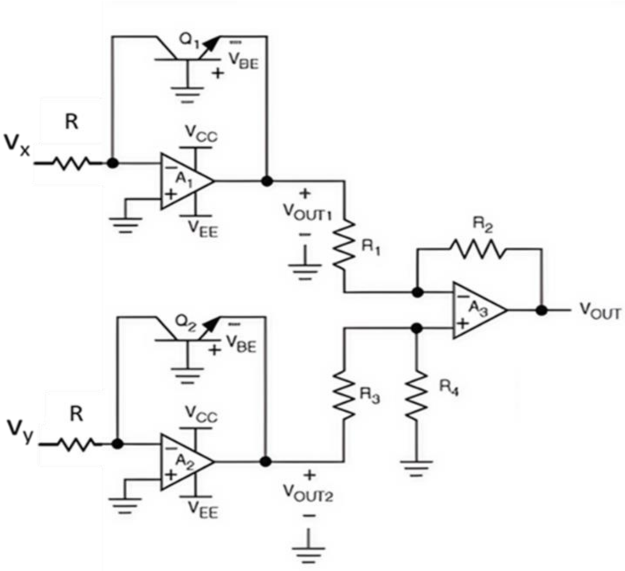


Fig. 8

15. The timer circuit in **Fig. 9** will flash the LED such that its ON time will be 3 seconds and OFF time will be 1 second. Assuming C = 10 μF, determine the value of R_1 and R_2.

A. R_1 = 6.26 MΩ, R_2 = 2.44 MΩ   B. R_1 = 3.66 MΩ, R_2 = 6.88 MΩ

C. R_1 = 2.88 MΩ, R_2 = 1.44 MΩ   D. R_1 = 4.76 MΩ, R_2 = 8.54 MΩ

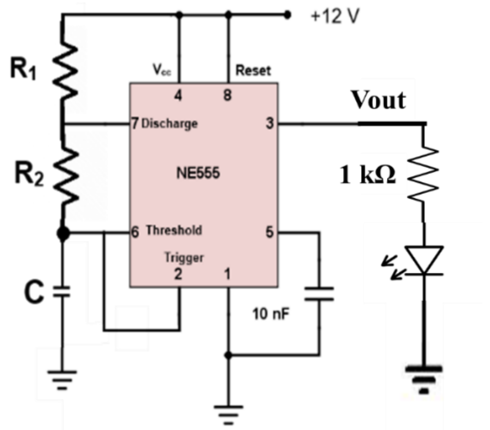


Fig. 9.

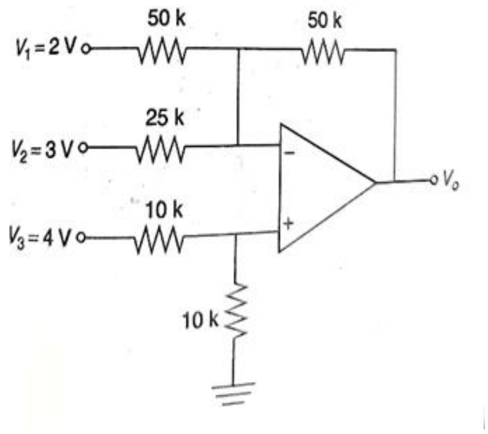


Fig. 10

16. Determine the output voltage (Vo) in the circuit shown in **Fig. 10**.

A. 0 B. - 2 V C. - 8V D. 10V

17. Determine the nature of feedback in the circuit shown in **Fig. 11**

A. Voltage shunt B. Voltage Series C. Current series **D. Current shunt**

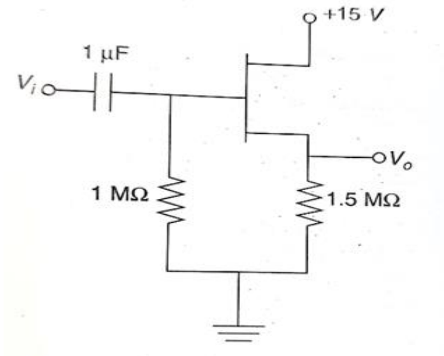


Fig. 11

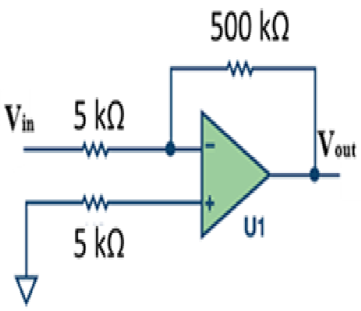


Fig. 12

18. Calculate the total output offset voltage if the input offset voltage & current in the circuit shown in **Fig. 12,** are 2 mV and 150 nA.

A. 277 mV B. 200 mV C. 0.75 mV D. 2.75 mV

19. In the VCCS circuit shown in **Fig. 13**, the expression of i_L will be

A. V_I / R_L B. $5V_I / R_L$ **C. $101V_I / R$** D. $100V_I / R$

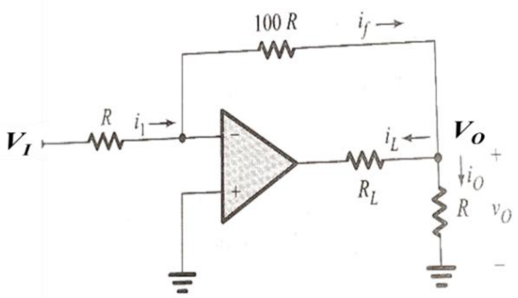


Fig. 13

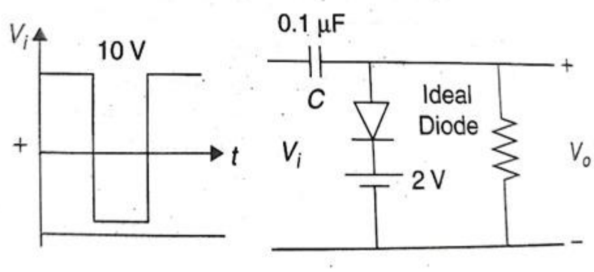
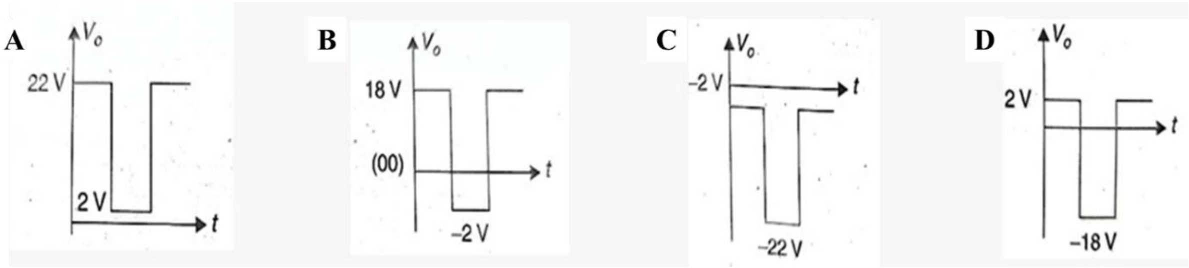


Fig. 14

20. Select the correct output V_o for given V_i in the clamping circuit shown in **Fig. 14**.

Ans: D



*****************End of Question******************

Self-Declaration:

I declare that I am not carrying with me:

1. Any written material on paper, clothes, or body parts
2. any mobile phone, communication or data storage devices.

Name:

Signature with date: